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PI Name:	Shay, Jerry W. Ph.D.		
Project Title:	Mouse Models of Cancer Risk and Prevention from Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
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Task Description:	Overall hypothesis: Low-dose radiation induces molecular manifestations of a pro-inflammatory response as a function of radiation type, radiation doses, doses rates, LET (linear energy transfer) value, and time. An oral available anti-inflammatory countermeasure, already in human clinical trials with a good safety profile, will significantly reduce proton and HZE-ion (high charge energy-ion) exposure associated tumor initiation and progression. The overarching hypothesis for this project is that space radiation induces molecular manifestations of a pro-inflammatory response as a function of radiation type, radiation doses, doses rates, LET value, and time. We are testing if an oral available anti-oxidant and anti-inflammatory countermeasure, already in human clinical trials with a good safety profile, CDDO, significantly reduces proton and HZE-ion exposure associated tumor initiation and progression. Based on experiments conducted at the NASA Space Radiation Laboratory (Brookhaven, NY) we demonstrate that HZE ion components of GCR (galactic cosmic radiation) result in persistent DNA damage and inflammatory signaling, increased mutations in tumor suppressor genes, and higher rates of cancer initiation and progression compared to that seen with similar doses of terrestrial radiation. While physical shielding may reduce some of the risks of space radiation, there is substantial evidence that biological countermeasures will be required to ensure that the established safety limits of increased lifetime fatal cancer risks are not exceeded. We are conducting GCR simulations consisting of fast switching between protons, helium, and silicon using a dose rate of 0.5 cGy/min and a total combined dose of between 27-30 cGy to more closely mimic the space environment on a trip to Mars and back. Finally, we are conducting experiments with the official NASA GCRsim with acute and protracted mixed fields. References Kim, S.B., Bozeman, R.G., Kaisani, A., Kim, W., Zhang, L., Richardson, J.A., Wright, W.E., and Shay, J.W. Radiation promotes colorectal cancer initiation and progression by inducing senescence-associated inflammatory responses. Oncogene DOI: 10.1038/onc2015.395, 2015. Norbury, J.W., Schimmerling, W., Slaba, T.C., Edouard Azzam, Francis F. Badavi, Giorgio Baiocco, Eric Benton, Veronica Bindi, Eleanor A. Blakely, Steve R. Blattnig, David A. Boothman, Thomas B. Borak, Richard A. Britten, Stan Curtis, Michael Dingfelder, Marco Durante, William Dynan, Amelia Eisch, S. Robin Elgart, Dudley T. Goodhead, Peter M. Guida, Lawrence H. Heilbronn, Christine E. Hellweg, Janice L. Huff, Amy Kronenberg, Chiara La Tessa, Derek Lowenstein, Jack Miller, Taksahi Morita, Livio Narici, Gregory A. Nelson, Ryan B. Norman, Takeo Ohnishi, Andrea Ottolenghi, Zarana S. Patel, Guenther Reitz, Adam Rusek, Ann-Sofie Schreurs, Lisa A. Scott-Carnell, Edward Semones, Jerry W. Shay, Vyacheslav A. Shurshakov, Lembit Sihver, Lisa C. Simonsen, Michael Story, Mitchell S. Turker, Yukio Uchihori, Jacqueline Williams, Cary J. Zeitlin. Galactic cosmic ray simulation at the NASA Space Radiation Laboratory. Life Sciences in Space Research 8:38-51, 2016. PMID: 26948012 Lutiel, K. Bozeman, R., Kaisani, A. Kim, S.B., Barron, S., Richardson, J.A., Shay, J.W. Proton radiation-induced cancer progression. Life Sciences in Space Research, in press, 2018. https://
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although radiation therapy is commonly used for treatment of many human diseases, including cancer, ionizing radiation produces reactive oxygen species that can damage both cancer and healthy cells in tissues. We have demonstrated using the biological countermeasure, CDDO, an anti-oxidant, anti-inflammatory modulator with a known mechanism of action, a reduction in both cancer initiation and progression in mouse models after exposure to either terrestrial or space radiation. We also demonstrated that CDDO can be used as a radioprotector in normal non cancerous human lung and breast epithelial cells exposed to space and terrestrial irradiations while cancer cells were not protected. This suggests the use of this oral available, non-toxic class of drug can protect non-cancerous healthy cells during radiotherapy, resulting in better outcomes with less toxicity for patients. CDDO is currently in a phase 3 clinical trial for patients with pulmonary arterial hypertension and Alport's syndrome. In preliminary experiments aspirin did not provide this protection. Other potential countermeasures such as metformin will be tested in the future. In the future, CDDO may be used to protect astronauts on long-term mission to Mars as well as patients receiving radiotherapy on Earth. This radioprotector may also have utility in protecting first responders to nuclear accidents.
Task Progress:	With the high energy and control upgrades at the NASA Space Radiation Laboratory (NSRL), Brookhaven, NY, experiments can now be conducted to better simulate the deep space environment that would occur on a Mars mission. During NSRL18B (June, 2018) an acute 0.5 Gy total exposure to the newly developed GCR simulation was initiated and first results from this acute exposure GCR simulation will be presented. During NSRL18C (October-November, 2018) a chronic/protracted 0.5 Gy total exposure was approved to better simulate the low dose rates expected in the deep space environment. This will consist of 24 separate days of irradiation (6-days per week) with a continuous exposure to background protons and helium and a sporadic heavy ion exposure. The acute experiments from NSRL18B consisted of the following • Wild type (WT), n=30 and lung cancer susceptible mice (LA1 mutant) on a 129 background (n=83), mixed male and females, 8-12 weeks old • 18 WT mice received 500 mGy GCRsim, 12 unirradiated • 63 lung cancer susceptible mice received 500mGy GCRsim, 20 unirradiated • Acute experiments, ~2 hrs, 0.5 cGy/min • First end points: 100 days +/- mitigator (CDDO, initiated two weeks post-irradiation for 30 days) • 100-day assays: weight, inflammation, oxyblot (DNP) oxidized proteins, 8-oxo-dG for number of DNA lesions, plasma lipid peroxidation (MDA assay), initial histopathology with tumor size and number • 365-day assays (to be completed): tumor grade, median, and overall survival In summary, the testing of acute and chronic exposures to a GCR simulation should determine if increases in more lethal cancer risks observed in acute GCR exposure are increased, decreased, or unchanged when similar overall doses are provided with low dose rates. This should determine if countermeasures will be required for astronauts to travel to Mars and back safely. The late biological effects from mixtures of the ions present in Galactic cosmic rays (GCR) such as protons, alpha

	particles, and high charge (Z) and energy E (HZE) particles are poorly understood, and at the present time astronaut's exposure to such radiation during exploration class missions represents an unacceptable level of cancer risk. Most of the work on understanding space radiation-induced cancer progression has been performed using mono-energetic single-ion beams. However, the space radiation environment consists of a wide variety of ion species with various ranges of energies. Using the fast beam switching technology recently developed in the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL), we can rapidly switch ion species allowing us to use mixed ions with different energies to simulate the actual radiation environment found in space. During the past year we exposed the lung cancer susceptible mouse model (K-rasLA-1) at the NSRL with three ion beams: Proton (H) (120 MeV/n) 20 cGy, Helium (He) (250 MeV/n) 5 cGy, and Silicon (Si) (300 MeV/n) 5 cGy with a dose rate of 0.5 cGy/min. Using three ion beams we performed whole body irradiations with a total dose of 30 cGy in two different orders: 3B-1 (H›He›Si) and 3B-2 (Si›He›H) and used 30 cGy H as a reference. In this study using the K-rasLA-1 mouse model, we show that whole-body irradiation with 3B-1 increases the incidence of cancer initiation and systemic oxidative stress in mice 100 days post-irradiation compared to 3B-2 and H only irradiation. Additionally, we observed an increase in adenomas with atypia and adenocarcinomas in 3B-1 irradiated mice but not in 3B-2 and H only irradiated mice. We also found that a non-toxic anti-inflammatory, anti-oxidative radioprotector (CDDO) provided 3 days prior to irradiation and one day post-irradiation reduced 3B-1 induced oxidative stress and cancer initiation almost back to baseline 100 days post irradiation. Thus, exposure to 3B-1 elicits significant changes in colon and lung cancer initiation that can be mitigated using CDDO.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Abstracts for Journals and Proceedings	Shay JW. "Mouse Models of Cancer Risk and Prevention from GCR Simulations." Presented at the 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Articles in Peer-reviewed Journals	Luitel K, Bozeman RG, Kaisani A, Kim SB, Barron S, Richardson JA, Shay JW. "Proton radiation-induced cancer progression." Life Sciences in Space Research. 2018 Nov;19:31-42. Epub 2018 Aug 18. https://doi.org/10.1016/j.lssr.2018.08.002 ; PubMed PMID: 30482279 , Nov-2018